

Data Curation and Management Plans

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2024/2025 Academic Year
Lecture Note 2

Outline

- 1 Selection and Appraisal
- 2 Data Archiving
- 3 Data Warehouse
- 4 Data preprocessing
- 5 Data Cleaning
- 6 Repository
- 7 Economics of data: preservation, access, and sustainability

Selection and Appraisal

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- There is the need to also understand the best criteria to use in making a choice in terms of data to accept and archive.
- Making a choice on the data to archive would require you to answer the following questions.

Selection and Appraisal

- Does the data fits into the mission and vision of the repository?
- Does the data fits into the research focus of the institution?
- What is the value of the data to the field (e.g. Social Sciences, Psychology, Medicine, Public Health etc.) of study the repository seek to archive its data
- How unique is the data and can it survive the test of time?
- Is the data useful in terms of using current and emerging research and statistical techniques?

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 - Data that can support both qualitative and quantitative analyses; data resulting from concurrent (both at the same time), sequential (one following the other), or conversion (one method to the other) mixed method study designs.

Selection and Appraisal

- Data that is interdisciplinary in nature
 - Data from interdisciplinary studies, and data resulting from studies using the research methods of multiple disciplines.

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- The data are in a format that facilitates ease of use.

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- You should be able to review the submission before making it available online.

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- Hence data archiving aids in the maximisation of invested resources in data production when there exist some form of expectation that they could be used in the future
- It provides an ongoing access to cultural and historical heritage of a nation.

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 - The unit responsible for coordinating the archiving system as well as managing the content packages.

Good Practices for Data Archiving

Integrity Feature	Related Good Practice
Content: Ensures that essential elements of digital content are preserved	An organisation is expected to explicitly identify and actively manage data to be preserved
Fixity: requires that changes to content are recorded, ideally from the moment of creation onward	At minimum, this feature might be addressed through routine use of a checksum (a computed value generated by widely-available utilities that uniquely identify a digital file) to detect intentional or unintentional changes to data and notify data managers for action
Reference: ensures content is uniquely and specifically identifiable in relation to other content across time	For example, an organisation is required to adopt and maintain a persistent identifier approach – a system for assigning and managing enduring identifiers that allows digital objects to be consistently and uniquely referred to over time
Context: documents and manages relationships of digital content	An organisation that preserves data document relationships between its own digital content and, to the extent possible, to data managed by other organisations

Process of Data Archiving at C-DAMAA

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- It summarises the processes of accepting, processing and putting the data out there for re-use.

DATA WAREHOUSE

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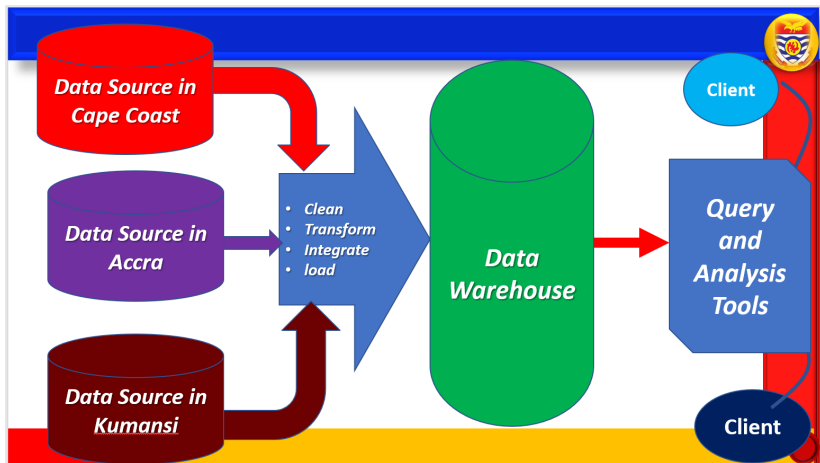
Data Warehouse

- A data warehouse is constructed by integrating data from multiple heterogeneous sources that support analytical reporting, and structured decision making.
- Data warehousing involves data cleaning, data integration, and data consolidations.
- A Data Warehouse works as a central repository where information arrives from one or more data sources.
- Data flows into a data warehouse from the transactional system and other relational databases.

DATA WAREHOUSES

- Data Warehouses:
 - Data spread in several databases –physically located at numerous sites
 - Data warehouse – repository of multiple DBs in single schema; resides at single site.
- Data warehousing processes
 - Data Cleaning
 - Data Integration
 - Data Transformation
 - Data Loading
 - Periodic data refreshing

DATA WAREHOUSE



DATA WAREHOUSES

- Data cleaning
 - Data Cleaning includes, filling in missing values, smoothing noisy data, identifying or removing outliers, and resolving inconsistencies.
- Data integration
 - Data Integration includes integration of multiple databases, data cubes, or files.
- Data transformation
 - State the scope of what data will be curated and establish written policies
- Clearly Stated Targeted Audience
 - Convert data from legacy or host format to warehouse format.
 - e.,g. user needs, assessment, gap analysis of existing services, stakeholder analysis.

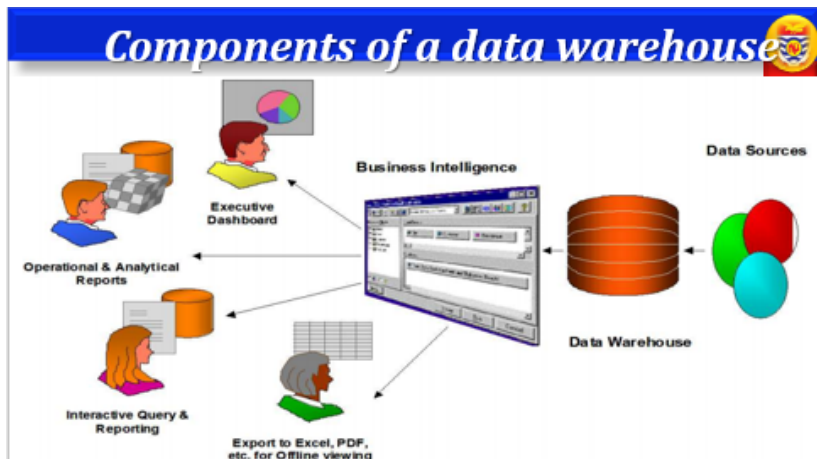
DATA WAREHOUSES

- Load
 - sort
 - summarize,
 - consolidate;
 - compute views;
 - check integrity.
 - Build indices and partitions.

DATA WAREHOUSES

- Data in a data warehouse are organized around major subjects.
- Data provide information on historical perspective
 - summarized on periodic dimension.
- Eg. Sales of an item for a region in a period
- Data warehouse model – multidimensional database structure / data cube
- Dimensions – Attributes / set of attributes
- Facts – Aggregated measures (Count /Sales amount).

DATA WAREHOUSES



Data preprocessing

- Why preprocess the data?
 - Data in the real world is dirty
 - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., occupation=""
 - noisy: containing errors or outliers
 - e.g., Salary="-10"
 - inconsistent: containing discrepancies in codes or names
 - e.g., Age="42" Birthday="03/07/1997"
 - e.g., Was rating "1,2,3", now rating "A, B, C"
 - e.g., discrepancy between duplicate records

Why Is Data Dirty?

- Incomplete data comes from.
 - n/a data value when collected
 - different consideration between the time when the data was collected and when it is analyzed.
 - human/hardware/software problems
- Noisy data comes from the process of data
 - collection
 - entry
 - transmission
- Inconsistent data comes from
 - Different data sources
 - Functional dependency violation

Major Tasks in Data Preprocessing

- Data cleaning .
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
 - Integration of multiple databases, data cubes, or files
- Data transformation
 - Normalization and aggregation
- Data reduction
 - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization
 - Part of data reduction but with particular importance, especially for numerical data

Data Cleaning

- Importance.
 - “Data cleaning is one of the three biggest problems in data warehousing”—Ralph Kimball
 - “Data cleaning is the number one problem in data warehousing”—DCI survey
- Data cleaning tasks
 - Fill in missing values
 - Identify outliers and smooth out noisy data
 - Correct inconsistent data
 - Resolve redundancy caused by data integration

Assignment

ASSIGNMENT

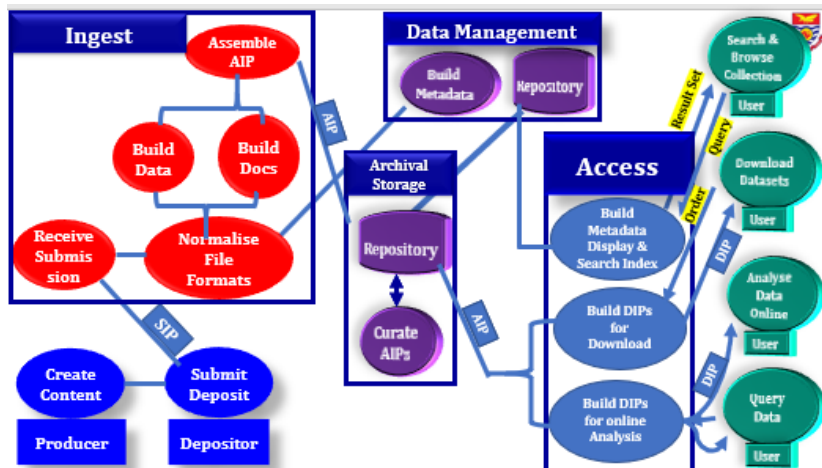
- Outline six (6) Data quality performance indicators recommendable for meeting the vision of your institution. (One Page)
- Identify and justify five administrative data points your organization should prioritise for archiving for the next five years. (1 Page)

Missing Data

- Data is not always available
 - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to .
 - equipment malfunction
 - inconsistent with other recorded data and thus deleted
 - data not entered due to misunderstanding
 - certain data may not be considered important at the time of entry
 - not register history or changes of the data
- Missing data may need to be inferred.

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (assuming the tasks in classification—not effective when the percentage of missing values per attribute varies considerably.
- Fill in the missing value manually: tedious + infeasible?.
- Fill in it automatically with
 - a global constant : e.g., “unknown”, a new class?!
 - the attribute mean
 - the attribute mean for all samples belonging to the same class: smarter
 - **the most probable value: inference-based such as Bayesian formula or decision tree**



Data Archiving System Functions

- Ingest
 - enables secure acceptance and quality control of data for preservation.
- Archival Storage
 - ensures the secure storage, management, and retrieval of preserved data
- Data Management
 - supports ongoing accumulation and availability of administrative data on the operation of and documentation of the content of an archive
- Administration
 - develops, maintains, and applies policies and procedures to operate and coordinate archival system functions

Data Archiving System Functions

- Preservation Planning
 - develops and recommends standards, policies, procedures and means for preserving digital content
- Access
 - finds and delivers content in an archive to authorised users
- Common Services
 - refers to the operating system, network services and security services needed to implement and manage a preservation repository

Information Packages

- Submission Information Package (SIP):
 - the state (format and structure) of the digital content when it is provided to the repository or unit responsible for preservation.
 - Effective preservation should make it possible always to be able to demonstrate the state of data as submitted (the original)
- Archival Information Package (AIP)
 - the state of digital content when it is placed into archival storage for preservation
 - An AIP accumulates information on preservation activities (e.g. data are migrated to a new format, preservation metadata are updated) so anything that might affect preservation or use of data is known and documented
- Dissemination Information Package (DIP):
 - the state of the digital content when it is made available to users over time

Data Repository

- A repository is a real time database that consolidates data from a variety sources to present a unified view of a single phenomenon.
- It is optimised to allow researchers to retrieve data from variety of issues for research.

Need of Data Repository

- Key issues faced by researchers today include:
- Lack of uniformity in data collection process.
- Duplication of data related to the same issue
- In ability to track progress and performance over time
- Manual reconciliation of data.

Data Repository Implementation- Challenges

- Storage capacity
- Computing power
- Reliability
- Accessibility of the data
- Electronic interface between all the ancillary data sources and the repository
- Network connection
- User interface

Economics of data: preservation, access, and sustainability

- Since with discussed issues of preservation and accessibility extensively, we focus much on sustainability of repositories.
- Sustainability is the capacity to generate or gain access to the resources, needed to protect and increase the value of the content or service for data repository.
- The process of data curation and management plans for the purposes of data archiving can only be effective, when preservation is seen as a sustainable economic activity.

Economics of data: preservation, access, and sustainability

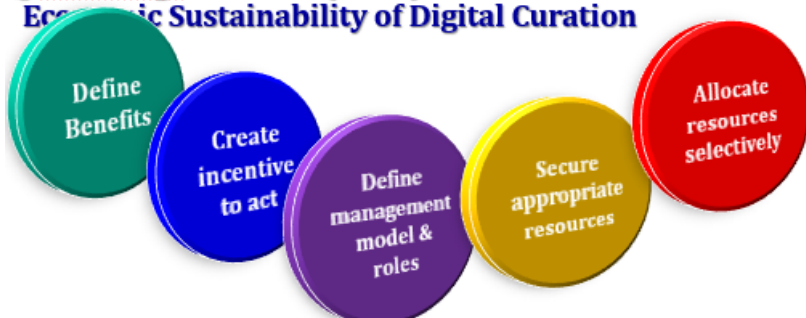
- The ability of any data repository to survive the test of time largely depend on the level of funding available for that purpose.
- Funding for the purposes data preservation continue to be a major problem. This has been attributed to:
- Starting a data repository is a capital intensive venture.
- In most cases people do not see any value coming from the preservation.

Economics of data: preservation, access, and sustainability

- Sustainable economics of data preservation is not just about finding more money.
- Its about establishing an economic activity that is firmly rooted in a compelling value preposition, clear incentives to act and a well-define roles and responsibilities (Kilbride, 2013).

Preservation as a sustainable economic activity

Rusbridge & Lavoie (2011) Draft Reference Model for Economic Sustainability of Digital Curation



Benefits and Incentives

Build case for preserving research data in institution

Establish benefits

- Outcomes that will be possible as a result of activity
- Objectives that won't be achievable if not performed



Identify key stakeholders

- Establish role in organisation
- Relation to research data preservation



Tailor message

- Target their "self interests"
- Encourage them to invest & accept responsibility

Establishing buy-in: Incentives

